CAT Quantitative Aptitude Topicwise PYQ Index (2017-2025)

A source-linked practice index covering 612 questions across CAT 2017-2025. Topics are ranked from most frequent to least frequent, then grouped by year and slot.

612 Questions	27 Topic groups	9 Exam years	12 Review flags

How to use: open each linked question, solve it under a timer, and record the result in the companion workbook. Difficulty marked with an asterisk is derived from public archive attempt statistics when available; Unrated means no attempt statistic was available. Topic labels are a practical preparation taxonomy. Rows marked for review are broad or image-dependent classifications that should be checked while solving.

Source archive: https://online.2iim.com/CAT-question-paper/

Priority topic map

Topics are sorted by question frequency, highest first. Counts represent individual questions, not sets.

Number System	Factors, Divisibility & Integers	**88**
Arithmetic	Percentages	**60**
Algebra	Logarithms, Exponents & Surds	**50**
Arithmetic	Ratios, Proportion & Variation	**44**
Arithmetic	Time, Speed & Distance	**44**
Geometry	Triangles, Quadrilaterals & Polygons	**40**
Arithmetic	Averages	**35**
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Algebra	Functions	**28**
Algebra	Sequences & Series	**22**
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Arithmetic	Simple & Compound Interest	**19**
Arithmetic	Time & Work	**14**
Modern Math	Permutations & Combinations	**14**
Algebra	Linear & Quadratic Equations	**12**
Arithmetic	Mixtures & Alligation	**12**
Geometry	Coordinate Geometry	**12**
Other	Mixed / Needs Review	**12**
Algebra	Inequalities & Optimization	**11**
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Number System	Factorials	**1**
Number System	Fractions & Decimals	**1**

NUMBER SYSTEM

Factors, Divisibility & Integers

88 questions | 2017: 7 | 2018: 9 | 2019: 10 | 2020: 12 | 2021: 7 | 2022: 12 | 2023: 12 | 2024: 7 | 2025: 12

CAT 2017 - 7 questions

1	26	CAT2017-S1-QA-Q26	Unrated	For how many integers n, will the inequality (n – 5) (n ...	Open
1	27	CAT2017-S1-QA-Q27	Unrated	If f 1 (x) = x 2 + 11x + n and ...	Open
1	28	CAT2017-S1-QA-Q28	Unrated	If a, b, c, and d are integers such that a + ...	Open
1	34	CAT2017-S1-QA-Q34	Unrated	Let a 1 , a 2 ,.....a 3n be an arithmetic progression ...	Open
2	21	CAT2017-S2-QA-Q21	Unrated	If the product of three consecutive positive integers is 15600 then the ...	Open
2	28	CAT2017-S2-QA-Q28	Unrated	How many different pairs (a, b) of positive integers are there such ...	Open
2	31	CAT2017-S2-QA-Q31	Unrated	If f(ab) = f(a)f(b) for all positive integers a and b, then ...	Open

CAT 2018 - 9 questions

1	3	CAT2018-S1-QA-Q03	Unrated	If f(x + 2) = f(x) + f(x + 1) for all ...	Open
1	10	CAT2018-S1-QA-Q10	Unrated	While multiplying three real numbers, Ashok took one of the numbers as ...	Open
1	21	CAT2018-S1-QA-Q21	Unrated	The number of integers x such that 0.25 < 2 x < ...	Open
2	5	CAT2018-S2-QA-Q05	Unrated	Let t 1 , t 2 ,… be real numbers such that ...	Open
2	7	CAT2018-S2-QA-Q07	Unrated	If N and x are positive integers such that N N = ...	Open
2	13	CAT2018-S2-QA-Q13	Unrated	The smallest integer n for which 4 n □ 17 19 holds, ...	Open
2	14	CAT2018-S2-QA-Q14	Unrated	The smallest integer n such that n 3 - 11n 2 + ...	Open
2	18	CAT2018-S2-QA-Q18	Unrated	If the sum of squares of two numbers is 97, then which ...	Open
2	23	CAT2018-S2-QA-Q23	Unrated	If a and b are integers such that 2x 2 − ax ...	Open

CAT 2019 - 10 questions

1	9	CAT2019-S1-QA-Q09	Unrated	For any positive integer n, let f(n) = n(n + 1) if ...	Open
1	24	CAT2019-S1-QA-Q24	Unrated	If m and n are integers such that (\\sqrt{2})) 19 3 4 ...	Open
1	33	CAT2019-S1-QA-Q33	Unrated	Consider a function f(x+y) = f(x) f(y) where x , y are ...	Open
2	3	CAT2019-S2-QA-Q03	Unrated	Let a, b, x, y be real numbers such that a 2 ...	Open
2	5	CAT2019-S2-QA-Q05	Unrated	Let a 1 , a 2 be integers such that a 1 ...	Open
2	6	CAT2019-S2-QA-Q06	Unrated	How many factors of 2 4 × 3 5 × 10 4 ...	Open
2	8	CAT2019-S2-QA-Q08	Unrated	What is the largest positive integer such that \\frac{n^2+7n+12}{n^2-n-12}) is also positive ...	Open
2	10	CAT2019-S2-QA-Q10	Unrated	Let f be a function such that f (mn) = f (m) ...	Open
2	22	CAT2019-S2-QA-Q22	Unrated	How many pairs (m,n) of positive integers satisfy the equation the equation ...	Open
2	25	CAT2019-S2-QA-Q25	Unrated	The quadratic equation x 2 + bx + c = 0 has ...	Open

CAT 2020 - 12 questions

1	15	CAT2020-S1-QA-Q15	Unrated	How many distinct positive integer-valued solutions exist to the equation (x 2 ...	Open

1	20	CAT2020-S1-QA-Q20	Unrated	If a, b and c are positive integers such that ab = ...	Open
2	4	CAT2020-S2-QA-Q04	Unrated	The number of pairs of integers(x,y) satisfying x ≥ y ≥ -20 ...	Open
2	12	CAT2020-S2-QA-Q12	Unrated	If x and y are non-negative integers such that x + 9 ...	Open
2	17	CAT2020-S2-QA-Q17	Unrated	The number of integers that satisfy the equality (x 2 - 5x ...	Open
2	18	CAT2020-S2-QA-Q18	Unrated	In how many ways can a pair of integers (x , a) ...	Open
3	1	CAT2020-S3-QA-Q01	Unrated	If x 1 = -1 and x m = x m + ...	Open
3	2	CAT2020-S3-QA-Q02	Unrated	Let N, x and y be positive integers such that N = ...	Open
3	7	CAT2020-S3-QA-Q07	Unrated	How many of the integers 1, 2, … , 120, are divisible ...	Open
3	12	CAT2020-S3-QA-Q12	Unrated	Let m and n be natural numbers such that n is even ...	Open
3	23	CAT2020-S3-QA-Q23	Unrated	Let m and n be positive integers, If x 2 + mx ...	Open
3	26	CAT2020-S3-QA-Q26	Unrated	How many pairs (a,b) of positive integers are there such that a ...	Open

CAT 2021 - 7 questions

1	2	CAT2021-S1-QA-Q02	Difficult	The number of integers n that satisfy the inequalities |n - 60| ...	Open
1	18	CAT2021-S1-QA-Q18	Difficult	The natural numbers are divided into groups as (1), (2, 3, 4), ...	Open
2	10	CAT2021-S2-QA-Q10	Difficult	Three positive integers x, y and z are in arithmetic progression. If ...	Open
2	15	CAT2021-S2-QA-Q15	Difficult	For all possible integers n satisfying 2.25 ≤ 2 + 2 n ...	Open
2	20	CAT2021-S2-QA-Q20	Difficult	For a sequence of real numbers x 1 , x 2 , ...	Open
3	18	CAT2021-S3-QA-Q18	Difficult	The number of distinct pairs of integers (m, n) satisfying |1+mn| < ...	Open
3	20	CAT2021-S3-QA-Q20	Difficult	If \(n\) is a positive integer such that \((\sqrt[7]{10})(\sqrt[7]{10})^{2} \ldots(\sqrt[7]{10})^{n}>999\), then the ...	Open

CAT 2022 - 12 questions

1	4	CAT2022-S1-QA-Q04	Unrated	For natural numbers \(x, y\), and \(z\), if \(x y+y z=19\) and ...	Open
1	5	CAT2022-S1-QA-Q05	Unrated	Let \(a, b, c\) be non-zero real numbers such that \(b^2 \lt ...	Open
1	12	CAT2022-S1-QA-Q12	Unrated	For any real number \(x\), let \([x]\) be the largest integer less ...	Open
1	15	CAT2022-S1-QA-Q15	Unrated	Let \(a\) and \(b\) be natural numbers. If \(a^2+a b+a=14\) and \(b^2+a ...	Open
1	18	CAT2022-S1-QA-Q18	Unrated	For any natural number \(n\), suppose the sum of the first \(n\) ...	Open
1	21	CAT2022-S1-QA-Q21	Unrated	Let \(A\) be the largest positive integer that divides all the numbers ...	Open
2	10	CAT2022-S2-QA-Q10	Unrated	Suppose for all integers \(x\), there are two functions \(f\) and \(g\) ...	Open
2	14	CAT2022-S2-QA-Q14	Unrated	The number of distinct integer values of \(n\) satisfying \(\frac{4-\log _2 n}{3-\log ...	Open
2	17	CAT2022-S2-QA-Q17	Unrated	The number of integer solutions of the equation \(\left(x^2-10\right)^{\left(x^2-3 x-10\right)}=1\) is	Open
3	1	CAT2022-S3-QA-Q01	Unrated	Suppose \(k\) is any integer such that the equation \(2 x^2+k x+5=0\) ...	Open
3	14	CAT2022-S3-QA-Q14	Unrated	If \((3+2 \sqrt{2})\) is a root of the equation \(a x^2+b x+c=0\), ...	Open
3	19	CAT2022-S3-QA-Q19	Unrated	A school has less than 5000 students and if the students are ...	Open

CAT 2023 - 12 questions

1	4	CAT2023-S1-QA-Q04	Unrated	Let \(n\) be the least positive integer such that 168 is a ...	Open
1	5	CAT2023-S1-QA-Q05	Unrated	The number of integer solutions of equation \(2|x|\left(x^2+1\right)=5 x^2 \) is	Open
1	7	CAT2023-S1-QA-Q07	Unrated	The equation \(x^3+(2 r+1) x^2+(4 r-1) x+2=0\) has -2 as one of ...	Open
2	1	CAT2023-S2-QA-Q01	Unrated	Let \(a, b, m\) and \(n\) be natural numbers such that \(a>1\) ...	Open
2	3	CAT2023-S2-QA-Q03	Unrated	For any natural numbers \(m, n\), and \(k\), such that \(k\) divides ...	Open
2	6	CAT2023-S2-QA-Q06	Unrated	The number of positive integers less than 50, having exactly two distinct ...	Open
2	7	CAT2023-S2-QA-Q07	Unrated	Let \(k\) be the largest integer such that the equation \((x-1)^2+2 k ...	Open
2	20	CAT2023-S2-QA-Q20	Unrated	Let both the series \(a_1, a_2, a_3, \ldots\) and \(b_1, b_2, b_3 ...	Open
3	2	CAT2023-S3-QA-Q02	Unrated	Let \(n\) and \(m\) be two positive integers such that there are ...	Open
3	6	CAT2023-S3-QA-Q06	Unrated	The sum of the first two natural numbers, each having 15 factors ...	Open
3	7	CAT2023-S3-QA-Q07	Unrated	Let \(n\) be any natural number such that \(5^{n-1} \lt 3^{n+1}\). Then, ...	Open
3	21	CAT2023-S3-QA-Q21	Unrated	Let \(a_n=46+8 n\) and \(b_n=98+4 n\) be two sequences for natural numbers ...	Open

CAT 2024 - 7 questions

1	18	CAT2024-S1-QA-Q18	Difficult	For any natural number \(n\), let \(a_n\) be the largest integer not ...	Open
1	22	CAT2024-S1-QA-Q22	Difficult	If \(x\) is a positive real number such that \(4 \log _{10} ...	Open
2	3	CAT2024-S2-QA-Q03	Difficult	If \(m\) and \(n\) are natural numbers such that \(n>1\), and \(m^n=2^{25} ...	Open
2	4	CAT2024-S2-QA-Q04	Difficult	A function \(f\) maps the set of natural numbers to whole numbers, ...	Open
2	9	CAT2024-S2-QA-Q09	Difficult	If \(a, b\) and \(c\) are positive real numbers such that \(a>10 ...	Open
3	1	CAT2024-S3-QA-Q01	Difficult	The number of distinct integer solutions \((x, y)\) of the equation \(|x+y|+|x-y|=2\), ...	Open
3	3	CAT2024-S3-QA-Q03	Moderate	If \((a+b \sqrt{3})^2=52+30 \sqrt{3}\), where \(a\) and \(b\) are natural numbers, then ...	Open

CAT 2025 - 12 questions

1	3	CAT2025-S1-QA-Q03	Difficult	The number of distinct integers \(n\) for which \(\log _{\left(\frac{1}{4}\right)}\left(n^2-7 n+11\right)>0\), is	Open
1	7	CAT2025-S1-QA-Q07	Difficult	The number of non-negative integer values of \(k\) for which the quadratic ...	Open
1	17	CAT2025-S1-QA-Q17	Difficult	The number of distinct pairs of integers \((x, y)\) satisfying the inequalities ...	Open
1	19	CAT2025-S1-QA-Q19	Difficult	For any natural number \(k\), let \(a_k=3^k\). The smallest natural number \(m\) ...	Open
1	22	CAT2025-S1-QA-Q22	Difficult	Let \(3 \leq x \leq 6\) and \(\left[x^2\right]=[x]^2\), where \([x]\) is the ...	Open
2	1	CAT2025-S2-QA-Q01	Difficult	If \(m \) and \(n \) are integers such that ...	Open
2	7	CAT2025-S2-QA-Q07	Difficult	If \(a , b , c \) and \(d \) ...	Open
2	9	CAT2025-S2-QA-Q09	Difficult	Suppose \(a , b , c \) are three distinct natural ...	Open
2	11	CAT2025-S2-QA-Q11	Difficult	The number of divisors of \(\left(2 ^ { 6 } ...	Open
2	20	CAT2025-S2-QA-Q20	Difficult	The sum of digits of the number \((625) ^ ...	Open

3	9	CAT2025-S3-QA-Q09	Difficult	If \(12 ^ { 12 x } \times 4 ^ { ...	Open
3	17	CAT2025-S3-QA-Q17	Difficult	The sum of all the digits of the number \(\left(10 ...	Open

ARITHMETIC

Percentages

60 questions | 2017: 7 | 2018: 7 | 2019: 13 | 2020: 5 | 2021: 5 | 2022: 2 | 2023: 7 | 2024: 7 | 2025: 7

CAT 2017 - 7 questions

1	1	CAT2017-S1-QA-Q01	Unrated	Arun's present age in years is 40% of Barun's. In another few ...	Open
1	5	CAT2017-S1-QA-Q05	Unrated	Ravi invests 50% of his monthly savings in fixed deposits. Thirty percent ...	Open
1	7	CAT2017-S1-QA-Q07	Unrated	A man travels by a motor boat down a river to his ...	Open
1	9	CAT2017-S1-QA-Q09	Unrated	The number of girls appearing for an admission test is twice the ...	Open
1	18	CAT2017-S1-QA-Q18	Unrated	A solid metallic cube is melted to form five solid cubes whose ...	Open
2	5	CAT2017-S2-QA-Q05	Unrated	Out of the shirts produced in a factory, 15% are defective, while ...	Open
2	10	CAT2017-S2-QA-Q10	Unrated	In a village, the production of food grains increased by 40% and ...	Open

CAT 2018 - 7 questions

1	8	CAT2018-S1-QA-Q08	Unrated	When they work alone, B needs 25% more time to finish a ...	Open
1	19	CAT2018-S1-QA-Q19	Unrated	In an examination, the maximum possible score is N while the pass ...	Open
1	24	CAT2018-S1-QA-Q24	Unrated	Points E, F, G, H lie on the sides AB, BC, CD, ...	Open
2	11	CAT2018-S2-QA-Q11	Unrated	A 20% ethanol solution is mixed with another ethanol solution, say, S ...	Open
2	19	CAT2018-S2-QA-Q19	Unrated	A jar contains a mixture of 175 ml water and 700 ml ...	Open
2	22	CAT2018-S2-QA-Q22	Unrated	Ramesh and Ganesh can together complete a work in 16 days. After ...	Open
2	33	CAT2018-S2-QA-Q33	Unrated	The strength of a salt solution is p% if 100 ml of ...	Open

CAT 2019 - 13 questions

1	1	CAT2019-S1-QA-Q01	Unrated	In a class, 60% of the students are girls and the rest ...	Open
1	6	CAT2019-S1-QA-Q06	Unrated	Meena scores 40% in an examination and after review, even though her ...	Open
1	14	CAT2019-S1-QA-Q14	Unrated	Two cars travel the same distance starting at 10:00 am and 11:00 ...	Open
1	18	CAT2019-S1-QA-Q18	Unrated	The income of Amala is 20% more than that of Bimala and ...	Open
1	22	CAT2019-S1-QA-Q22	Unrated	One can use three different transports which move at 10, 20, and ...	Open
1	25	CAT2019-S1-QA-Q25	Unrated	A chemist mixes two liquids 1 and 2. One litre of liquid ...	Open
2	9	CAT2019-S2-QA-Q09	Unrated	In 2010, a library contained a total of 11500 books in two ...	Open
2	15	CAT2019-S2-QA-Q15	Unrated	Anil alone can do a job in 20 days while Sunil alone ...	Open
2	17	CAT2019-S2-QA-Q17	Unrated	In an examination, the score of A was 10% less than that ...	Open
2	21	CAT2019-S2-QA-Q21	Unrated	Two ants A and B start from a point P on a ...	Open
2	23	CAT2019-S2-QA-Q23	Unrated	The salaries of Ramesh, Ganesh and Rajesh were in the ratio 6:5:7 ...	Open
2	30	CAT2019-S2-QA-Q30	Unrated	The strength of a salt solution is p% if 100 ml of ...	Open
2	34	CAT2019-S2-QA-Q34	Unrated	John gets Rs 57 per hour of regular work and Rs 114 ...	Open

CAT 2020 - 5 questions

1	4	CAT2020-S1-QA-Q04	Unrated	A train travelled at one-thirds of its usual speed, and hence reached ...	Open
1	23	CAT2020-S1-QA-Q23	Unrated	In a group of people, 28% of the members are young while ...	Open
2	23	CAT2020-S2-QA-Q23	Unrated	In May, John bought the same amount of rice and the same ...	Open
3	8	CAT2020-S3-QA-Q08	Unrated	In the final examination, Bishnu scored 52% and Asha scored 64%. The ...	Open
3	18	CAT2020-S3-QA-Q18	Unrated	Two alcohol solutions, A and B, are mixed in the proportion 1:3 ...	Open

CAT 2021 - 5 questions

1	11	CAT2021-S1-QA-Q11	Difficult	The strength of an indigo solution in percentage is equal to the ...	Open
2	7	CAT2021-S2-QA-Q07	Moderate	A box has 450 balls, each either white or black, there being ...	Open
3	1	CAT2021-S3-QA-Q01	Difficult	If a certain weight of an alloy of silver and copper is ...	Open
3	10	CAT2021-S3-QA-Q10	Difficult	The total of male and female populations in a city increased by ...	Open
3	14	CAT2021-S3-QA-Q14	Easy	In a tournament, a team has played 40 matches so far and ...	Open

CAT 2022 - 2 questions

2	11	CAT2022-S2-QA-Q11	Unrated	In an election, there were four candidates and 80% of the registered ...	Open
3	16	CAT2022-S3-QA-Q16	Unrated	A group of \(N\) people worked on a project. They finished \(35 ...	Open

CAT 2023 - 7 questions

1	9	CAT2023-S1-QA-Q09	Unrated	The minor angle between the hours hand and minutes hand of a ...	Open
1	12	CAT2023-S1-QA-Q12	Unrated	The salaries of three friends Sita, Gita and Mita are initially in ...	Open
2	8	CAT2023-S2-QA-Q08	Unrated	In a company, 20% of the employees work in the manufacturing department. ...	Open
3	9	CAT2023-S3-QA-Q09	Unrated	Anil mixes cocoa with sugar in the ratio 3 : 2 to ...	Open
3	10	CAT2023-S3-QA-Q10	Unrated	The population of a town in 2020 was 100000 . The population ...	Open
3	14	CAT2023-S3-QA-Q14	Unrated	A fruit seller has a stock of mangoes, bananas and apples with ...	Open
3	16	CAT2023-S3-QA-Q16	Unrated	Gautam and Suhani, working together, can finish a job in 20 days. ...	Open

CAT 2024 - 7 questions

1	6	CAT2024-S1-QA-Q06	Moderate	In September, the incomes of Kamal, Amal and Vimal are in the ...	Open
2	10	CAT2024-S2-QA-Q10	Difficult	A company has 40 employees whose names are listed in a certain ...	Open
2	12	CAT2024-S2-QA-Q12	Difficult	A fruit seller has a stock of mangoes, bananas and apples with ...	Open
2	19	CAT2024-S2-QA-Q19	Moderate	A vessel contained a certain amount of a solution of acid and ...	Open
3	5	CAT2024-S3-QA-Q05	Difficult	In a group of 250 students, the percentage of girls was at ...	Open
3	8	CAT2024-S3-QA-Q08	Easy	After two successive increments, Gopal's salary became 187.5% of his initial salary. ...	Open
3	10	CAT2024-S3-QA-Q10	Difficult	A certain amount of water was poured into a 300 litre container ...	Open

CAT 2025 - 7 questions

1	9	CAT2025-S1-QA-Q09	Difficult	A container holds 200 litres of a solution of acid and water, ...	Open
1	14	CAT2025-S1-QA-Q14	Difficult	In a class, there were more than 10 boys and a certain ...	Open
1	21	CAT2025-S1-QA-Q21	Difficult	Kamala divided her investment of Rs 100000 between stocks, bonds, and gold. ...	Open
2	17	CAT2025-S2-QA-Q17	Difficult	A mixture of coffee and cocoa, 16% of which is coffee, costs ...	Open
2	22	CAT2025-S2-QA-Q22	Easy	A certain amount of money was divided among Pinu, Meena, Rinu and ...	Open
3	12	CAT2025-S3-QA-Q12	Moderate	The average salary of 5 managers and 25 engineers in a company ...	Open
3	16	CAT2025-S3-QA-Q16	Difficult	Ankita walks from A to C through B, and runs back through ...	Open

ALGEBRA

Logarithms, Exponents & Surds

50 questions | 2017: 4 | 2018: 6 | 2019: 5 | 2020: 9 | 2021: 3 | 2022: 3 | 2023: 9 | 2024: 6 | 2025: 5

CAT 2017 - 4 questions

1	23	CAT2017-S1-QA-Q23	Unrated	The value of log 0.008 √5 + log √3 81 – 7 ...	Open
1	24	CAT2017-S1-QA-Q24	Unrated	If 9 2x – 1 – 81 x-1 = 1944, then x ...	Open
2	22	CAT2017-S2-QA-Q22	Unrated	If x is a real number such that log 3 5 = ...	Open
2	25	CAT2017-S2-QA-Q25	Unrated	If 9 x - (\\frac{1}{2})) – 2 2x – 2 = 4 ...	Open

CAT 2018 - 6 questions

1	11	CAT2018-S1-QA-Q11	Unrated	If x is a positive quantity such that 2 x = 3 ...	Open
1	12	CAT2018-S1-QA-Q12	Unrated	If log 12 81 = p, then 3(\\frac{4 - p}{4 + p})) ...	Open
1	25	CAT2018-S1-QA-Q25	Unrated	Given that x 2018 y 2017 = 1/2 and x 2016 y ...	Open
1	27	CAT2018-S1-QA-Q27	Unrated	If log 2 (5 + log 3 a) = 3 and log ...	Open
2	9	CAT2018-S2-QA-Q09	Unrated	If p 3 = q 4 = r 5 = s 6 ...	Open
2	34	CAT2018-S2-QA-Q34	Unrated	\\frac{1}{log_{2}100}\\) - \\frac{1}{log_{4}100}\\) + \\frac{1}{log_{5}100}\\) - \\frac{1}{log_{10}100}\\) + \\frac{1}{log_{20}100}\\) - \\frac{1}{log_{25}100}\\) + ...	Open

CAT 2019 - 5 questions

1	26	CAT2019-S1-QA-Q26	Unrated	Let x and y be positive real numbers such that log 5 ...	Open
2	1	CAT2019-S2-QA-Q01	Unrated	The real root of the equation 2 6x + 2 3x+2 - ...	Open
2	13	CAT2019-S2-QA-Q13	Unrated	Let A be a real number. Then the roots of the equation ...	Open
2	19	CAT2019-S2-QA-Q19	Unrated	If x is a real number ,then \\sqrt{log_{e}\frac{4x - x^2}{3}}) is a ...	Open
2	31	CAT2019-S2-QA-Q31	Unrated	If 5 x - 3 y = 13438 and 5 x-1 + ...	Open

CAT 2020 - 9 questions

1	5	CAT2020-S1-QA-Q05	Unrated	If log 4 5 = (log 4 y) (log 6 √5), then ...	Open
1	6	CAT2020-S1-QA-Q06	Unrated	The number of real-valued solutions of the equation 2 x + 2 ...	Open
1	9	CAT2020-S1-QA-Q09	Unrated	If x = (4096) 7+4√3 , then which of the following equals ...	Open
1	21	CAT2020-S1-QA-Q21	Unrated	If y is a negative number such that 2 y 2 log ...	Open
2	5	CAT2020-S2-QA-Q05	Unrated	The value of log a \\frac{a}{b}) + log b \\frac{b}{a}), for 1 ...	Open
2	10	CAT2020-S2-QA-Q10	Unrated	For real x, the maximum possible value of \\frac{x}{√(1 + x^{4})}) is	Open
3	3	CAT2020-S3-QA-Q03	Unrated	Let log a 30 = A, log a \\frac{5}{3}) = -B and ...	Open
3	10	CAT2020-S3-QA-Q10	Unrated	\\frac{2×4×8×16}{(log_{2}4)^{2}(log_{4}8)^{3}(log_{8}16)^{4}}) equals	Open
3	11	CAT2020-S3-QA-Q11	Unrated	If a,b,c are non-zero and 14 a = 36 b = 84 ...	Open

CAT 2021 - 3 questions

1	4	CAT2021-S1-QA-Q04	Difficult	If 5 - \\log _{10} \sqrt{1+x}) + 4 \\log _{10} \sqrt{1-x}) = ...	Open

2	2	CAT2021-S2-QA-Q02	Moderate	If log 2 [3 + log 3 {4 + log 4 (x ...	Open
3	6	CAT2021-S3-QA-Q06	Difficult	For a real number a, if \(\frac{\log _{15} a+\log _{32} a}{\left(\log _{15} ...	Open

CAT 2022 - 3 questions

2	9	CAT2022-S2-QA-Q09	Unrated	Let \(r\) and \(c\) be real numbers. If \(r\) and \(-r\) are ...	Open
3	2	CAT2022-S3-QA-Q02	Unrated	The minimum possible value of \(\frac{x^2-6 x+10}{3-x}\), for \(x\lt3\), is	Open
3	11	CAT2022-S3-QA-Q11	Unrated	If \(\left(\sqrt{\frac{7}{5}}\right)^{3 x-y}=\frac{875}{2401}\) and \(\left(\frac{4 a}{b}\right)^{6 x-y}=\left(\frac{2 a}{b}\right)^{y-6 x}\), for all non-zero ...	Open

CAT 2023 - 9 questions

1	1	CAT2023-S1-QA-Q01	Unrated	If \(x\) and \(y\) are positive real numbers such that \(\log _x\left(x^2+12\right)=4\) ...	Open
1	2	CAT2023-S1-QA-Q02	Unrated	If \(x\) and \(y\) are real numbers such that \(x^2+(x-2 y-1)^2=-4 y(x+y)\), ...	Open
1	6	CAT2023-S1-QA-Q06	Unrated	Let \(\alpha\) and \(\beta\) be the two distinct roots of the equation ...	Open
1	22	CAT2023-S1-QA-Q22	Unrated	A lab experiment measures the number of organisms at 8 am every ...	Open
2	2	CAT2023-S2-QA-Q02	Unrated	The sum of all possible values of \(x\) satisfying the equation \(2^{4 ...	Open
2	5	CAT2023-S2-QA-Q05	Unrated	For some positive real number \(x\), if \(\log _{\sqrt{3}}(x)+\frac{\log _x(25)}{\log _x(0.008)}=\frac{16}{3}\), then ...	Open
2	21	CAT2023-S2-QA-Q21	Unrated	If \(p^2+q^2-29=2 p q-20=52-2 p q\), then the difference between the maximum ...	Open
3	3	CAT2023-S3-QA-Q03	Unrated	For some real numbers \(a\) and \(b\), the system of equations \(x+y=4\) ...	Open
3	4	CAT2023-S3-QA-Q04	Unrated	If \(x\) is a positive real number such that \(x^8+\left(\frac{1}{x}\right)^8=47\), then the ...	Open

CAT 2024 - 6 questions

1	3	CAT2024-S1-QA-Q03	Difficult	The sum of all real values of \(k\) for which \(\left(\frac{1}{8}\right)^k \times\left(\frac{1}{32768}\right)^{\frac{1}{3}}=\frac{1}{8} ...	Open
1	4	CAT2024-S1-QA-Q04	Difficult	In the \(X Y\)-plane, the area, in sq. units, of the region ...	Open
1	8	CAT2024-S1-QA-Q08	Difficult	Let \(x, y\), and \(z\) be real numbers satisfying $$ \begin{aligned} & ...	Open
2	16	CAT2024-S2-QA-Q16	Difficult	The roots \(\alpha, \beta\) of the equation \(3 x^2+\lambda x-1=0\), satisfy \(\frac{1}{\alpha^2}+\frac{1}{\beta^2}=15\). ...	Open
3	16	CAT2024-S3-QA-Q16	Moderate	If \(3^a=4,4^b=5,5^c=6,6^d=7,7^e=8\) and \(8^f=9\), then the value of the product abcdef is	Open
3	22	CAT2024-S3-QA-Q22	Difficult	The sum of all distinct real values of \(x\) that satisfy the ...	Open

CAT 2025 - 5 questions

2	5	CAT2025-S2-QA-Q05	Difficult	If \(9 ^ { x ^ { 2 } + 2 ...	Open
2	14	CAT2025-S2-QA-Q14	Difficult	The set of all real values of \(x \) for which ...	Open
2	16	CAT2025-S2-QA-Q16	Difficult	The equations \(3 x ^ { 2 } - 5 x ...	Open

3	4	CAT2025-S3-QA-Q04	Difficult	If \(\left(x ^ { 2 } + \frac { 1 ...	Open
3	7	CAT2025-S3-QA-Q07	Difficult	The sum of all possible real values of \(x \) for ...	Open

ARITHMETIC

Ratios, Proportion & Variation

44 questions | 2017: 5 | 2018: 5 | 2019: 5 | 2020: 7 | 2021: 4 | 2022: 4 | 2023: 3 | 2024: 4 | 2025: 7

CAT 2017 - 5 questions

1	10	CAT2017-S1-QA-Q10	Unrated	A stall sells popcorn and chips in packets of three sizes: large, ...	Open
1	13	CAT2017-S1-QA-Q13	Unrated	If a and b are integers of opposite signs such that (a ...	Open
1	31	CAT2017-S1-QA-Q31	Unrated	If the square of the 7 th term of an arithmetic progression ...	Open
2	11	CAT2017-S2-QA-Q11	Unrated	If a, b, c are three positive integers such that a and ...	Open
2	13	CAT2017-S2-QA-Q13	Unrated	Amal can complete a job in 10 days and Bimal can complete ...	Open

CAT 2018 - 5 questions

1	20	CAT2018-S1-QA-Q20	Unrated	Let x, y, z be three positive real numbers in a geometric ...	Open
1	26	CAT2018-S1-QA-Q26	Unrated	Raju and Lalitha originally had marbles in the ratio 4 : 9. ...	Open
2	1	CAT2018-S2-QA-Q01	Unrated	Points A, P, Q and B lie on the same line such ...	Open
2	12	CAT2018-S2-QA-Q12	Unrated	The area of a rectangle and the square of its perimeter are ...	Open
2	24	CAT2018-S2-QA-Q24	Unrated	The scores of Amal and Bimal in an examination are in the ...	Open

CAT 2019 - 5 questions

1	7	CAT2019-S1-QA-Q07	Unrated	Corners are cut off from an equilateral triangle T to produce a ...	Open
1	12	CAT2019-S1-QA-Q12	Unrated	The product of two positive numbers is 616. If the ratio of ...	Open
1	23	CAT2019-S1-QA-Q23	Unrated	Amala, Bina, and Gouri invest money in the ratio 3 : 4 ...	Open
1	27	CAT2019-S1-QA-Q27	Unrated	If the rectangular faces of a brick have their diagonals in the ...	Open
2	16	CAT2019-S2-QA-Q16	Unrated	In an examination, Rama's score was one-twelfth of the sum of the ...	Open

CAT 2020 - 7 questions

1	18	CAT2020-S1-QA-Q18	Unrated	A circle is inscribed in a rhombus with diagonals 12 cm and ...	Open
1	24	CAT2020-S1-QA-Q24	Unrated	An alloy is prepared by mixing metals A, B, C in the ...	Open
2	6	CAT2020-S2-QA-Q06	Unrated	Let the m-th and n-th terms of a Geometric progression be \\frac{3}{4}) ...	Open
2	15	CAT2020-S2-QA-Q15	Unrated	The distance from B to C is thrice that from A to ...	Open
2	16	CAT2020-S2-QA-Q16	Unrated	The sum of perimeters of an equilateral triangle and a rectanmgle is ...	Open
2	20	CAT2020-S2-QA-Q20	Unrated	A and B are two points on a straight line. Ram runs ...	Open
2	25	CAT2020-S2-QA-Q25	Unrated	A sum of money is split among Amal, Sunil and Mita so ...	Open

CAT 2021 - 4 questions

1	17	CAT2021-S1-QA-Q17	Moderate	The amount Neeta and Geeta together earn in a day equals what ...	Open
2	5	CAT2021-S2-QA-Q05	Difficult	Two trains A and B were moving in opposite directions, their speeds ...	Open

3	2	CAT2021-S3-QA-Q02	Easy	Anil can paint a house in 12 days while Barun can paint ...	Open
3	17	CAT2021-S3-QA-Q17	Difficult	A tea shop offers tea in cups of three different sizes. The ...	Open

CAT 2022 - 4 questions

1	8	CAT2022-S1-QA-Q08	Unrated	Pinky is standing in a queue at a ticket counter. Suppose the ...	Open
1	13	CAT2022-S1-QA-Q13	Unrated	In a village, the ratio of number of males to females is ...	Open
2	2	CAT2022-S2-QA-Q02	Unrated	Working alone, the times taken by Anu, Tanu and Manu to complete ...	Open
2	3	CAT2022-S2-QA-Q03	Unrated	Regular polygons A and B have number of sides in the ratio ...	Open

CAT 2023 - 3 questions

2	18	CAT2023-S2-QA-Q18	Unrated	A triangle is drawn with its vertices on the circle \(C\) such ...	Open
3	15	CAT2023-S3-QA-Q15	Unrated	The number of coins collected per week by two coin-collectors A and ...	Open
3	18	CAT2023-S3-QA-Q18	Unrated	A rectangle with the largest possible area is drawn inside a semicircle ...	Open

CAT 2024 - 4 questions

1	1	CAT2024-S1-QA-Q01	Difficult	A fruit seller has a total of 187 fruits consisting of apples, ...	Open
2	18	CAT2024-S2-QA-Q18	Difficult	Three circles of equal radii touch (but not cross) each other externally. ...	Open
2	22	CAT2024-S2-QA-Q22	Difficult	When Rajesh's age was same as the present age of Garima, the ...	Open
3	19	CAT2024-S3-QA-Q19	Moderate	Rajesh and Vimal own 20 hectares and 30 hectares of agricultural land, ...	Open

CAT 2025 - 7 questions

1	11	CAT2025-S1-QA-Q11	Difficult	The ratio of the number of students in the morning shift and ...	Open
1	12	CAT2025-S1-QA-Q12	Difficult	In a circle with center \(C\) and radius \(6 \sqrt{2} \mathrm{~cm}, P ...	Open
2	6	CAT2025-S2-QA-Q06	Difficult	Let ABCDEF be a regular hexagon and P and Q be the ...	Open
2	13	CAT2025-S2-QA-Q13	Moderate	The ratio of expenditures of Lakshmi and Meenakshi is 2 : 3, ...	Open
3	8	CAT2025-S3-QA-Q08	Difficult	The ratio of the number of coins in boxes A and B ...	Open
3	19	CAT2025-S3-QA-Q19	Moderate	The rate of water flow through three pipes A, B and C ...	Open
3	22	CAT2025-S3-QA-Q22	Difficult	Vessels A and B contain 60 litres of alcohol and 60 litres ...	Open

ARITHMETIC

Time, Speed & Distance

44 questions | 2017: 9 | 2018: 7 | 2019: 4 | 2020: 8 | 2021: 2 | 2022: 5 | 2023: 3 | 2024: 3 | 2025: 3

CAT 2017 - 9 questions

1	4	CAT2017-S1-QA-Q04	Unrated	A man leaves his home and walks at a speed of 12 ...	Open
1	19	CAT2017-S1-QA-Q19	Unrated	A ball of diameter 4 cm is kept on top of a ...	Open
1	20	CAT2017-S1-QA-Q20	Unrated	Let ABC be a right-angled triangle with BC as the hypotenuse. Lengths ...	Open
1	30	CAT2017-S1-QA-Q30	Unrated	The shortest distance of the point (\\frac{1}{2}),1) from the curve y = ...	Open
2	2	CAT2017-S2-QA-Q02	Unrated	In a 10 km race. A, B, and C, each running at ...	Open
2	4	CAT2017-S2-QA-Q04	Unrated	Arun drove from home to his hostel at 60 miles per hour. ...	Open
2	12	CAT2017-S2-QA-Q12	Unrated	A motorbike leaves point A at 1 pm and moves towards point ...	Open
2	16	CAT2017-S2-QA-Q16	Unrated	The base of a vertical pillar with uniform cross section is a ...	Open
2	20	CAT2017-S2-QA-Q20	Unrated	Let P be an interior point of a right-angled isosceles triangle ABC ...	Open

CAT 2018 - 7 questions

1	4	CAT2018-S1-QA-Q04	Unrated	The distance from A to B is 60 km. Partha and Narayan ...	Open
1	17	CAT2018-S1-QA-Q17	Unrated	Point P lies between points A and B such that the length ...	Open
1	23	CAT2018-S1-QA-Q23	Unrated	Train T leaves station X for station Y at 3 pm. Train ...	Open
1	30	CAT2018-S1-QA-Q30	Unrated	In a circle, two parallel chords on the same side of a ...	Open
2	17	CAT2018-S2-QA-Q17	Unrated	On a long stretch of east-west road, A and B are two ...	Open
2	20	CAT2018-S2-QA-Q20	Unrated	Points A and B are 150 km apart. Cars 1 and 2 ...	Open
2	25	CAT2018-S2-QA-Q25	Unrated	A triangle ABC has area 32 sq units and its side BC, ...	Open

CAT 2019 - 4 questions

1	19	CAT2019-S1-QA-Q19	Unrated	In a race of three horses, the first beat the second by ...	Open
1	32	CAT2019-S1-QA-Q32	Unrated	The wheels of bicycles A and B have radii 30 cm and ...	Open
2	12	CAT2019-S2-QA-Q12	Unrated	A cyclist leaves A at 10 am and reaches B at 11 ...	Open
2	14	CAT2019-S2-QA-Q14	Unrated	John jogs on track A at 6 kmph and Mary jogs on ...	Open

CAT 2020 - 8 questions

1	7	CAT2020-S1-QA-Q07	Unrated	A straight road connects points A and B. Car 1 travels from ...	Open
1	14	CAT2020-S1-QA-Q14	Unrated	Two persons are walking beside a railway track at respective speeds of ...	Open
1	19	CAT2020-S1-QA-Q19	Unrated	Leaving home at the same time, Amal reaches office at 10:15 am ...	Open
2	1	CAT2020-S2-QA-Q01	Unrated	In a car race, car A beats car B by 45 km, ...	Open
2	19	CAT2020-S2-QA-Q19	Unrated	Two circular tracks T1 and T2 of radii 100 m and 20 ...	Open
3	13	CAT2020-S3-QA-Q13	Unrated	Anil, Sunil, and Ravi run along a circular path of length 3 ...	Open
3	21	CAT2020-S3-QA-Q21	Unrated	A and B are two railway stations 90 km apart. A train ...	Open

3	22	CAT2020-S3-QA-Q22	Unrated	Vimla starts for office every day at 9 am and reaches exactly ...	Open

CAT 2021 - 2 questions

1	21	CAT2021-S1-QA-Q21	Moderate	Two trains cross each other in 14 seconds when running in opposite ...	Open
3	3	CAT2021-S3-QA-Q03	Difficult	Mira and Amal walk along a circular track, starting from the same ...	Open

CAT 2022 - 5 questions

1	6	CAT2022-S1-QA-Q06	Unrated	Trains A and B start traveling at the same time towards each ...	Open
2	22	CAT2022-S2-QA-Q22	Unrated	Two ships meet mid-ocean, and then, one ship goes south and the ...	Open
3	10	CAT2022-S3-QA-Q10	Unrated	Two cars travel from different locations at constant speeds. To meet each ...	Open
3	12	CAT2022-S3-QA-Q12	Unrated	Moody takes 30 seconds to finish riding an escalator if he walks ...	Open
3	21	CAT2022-S3-QA-Q21	Unrated	Two ships are approaching a port along straight routes at constant speeds. ...	Open

CAT 2023 - 3 questions

1	14	CAT2023-S1-QA-Q14	Unrated	Arvind travels from town A to town B, and Surbhi from town ...	Open
2	11	CAT2023-S2-QA-Q11	Unrated	Ravi is driving at a speed of 40 km/h on a road. ...	Open
3	13	CAT2023-S3-QA-Q13	Unrated	A boat takes 2 hours to travel downstream a river from port ...	Open

CAT 2024 - 3 questions

1	15	CAT2024-S1-QA-Q15	Difficult	Two places A and B are 45 kms apart and connected by ...	Open
2	17	CAT2024-S2-QA-Q17	Difficult	A bus starts at 9 am and follows a fixed route every ...	Open
3	15	CAT2024-S3-QA-Q15	Moderate	A train travelled a certain distance at a uniform speed. Had the ...	Open

CAT 2025 - 3 questions

1	10	CAT2025-S1-QA-Q10	Difficult	Shruti travels a distance of 224 km in four parts for a ...	Open
2	19	CAT2025-S2-QA-Q19	Difficult	Rita and Sneha can row a boat at 5 km/h and 6 ...	Open
3	6	CAT2025-S3-QA-Q06	Difficult	Rahul starts on his journey at 5 pm at a constant speed ...	Open

GEOMETRY

Triangles, Quadrilaterals & Polygons

40 questions | 2017: 4 | 2018: 1 | 2019: 3 | 2020: 2 | 2021: 9 | 2022: 5 | 2023: 6 | 2024: 4 | 2025: 6

CAT 2017 - 4 questions

1	16	CAT2017-S1-QA-Q16	Unrated	From a triangle ABC with sides of lengths 40 ft, 25 ft ...	Open
2	1	CAT2017-S2-QA-Q01	Unrated	The numbers 1, 2,..., 9 are arranged in a 3 X 3 ...	Open
2	15	CAT2017-S2-QA-Q15	Unrated	Let ABCDEF be a regular hexagon with each side of length 1 ...	Open
2	19	CAT2017-S2-QA-Q19	Unrated	If three sides of a rectangular park have a total length 400 ...	Open

CAT 2018 - 1 questions

1	9	CAT2018-S1-QA-Q09	Unrated	Given an equilateral triangle T1 with side 24 cm, a second triangle ...	Open

CAT 2019 - 3 questions

2	4	CAT2019-S2-QA-Q04	Unrated	In a triangle ABC, medians AD and BE are perpendicular to each ...	Open
2	11	CAT2019-S2-QA-Q11	Unrated	Let A and B be two regular polygons having a and b ...	Open
2	20	CAT2019-S2-QA-Q20	Unrated	Let ABC be a right-angled triangle with hypotenuse BC of length 20 ...	Open

CAT 2020 - 2 questions

2	2	CAT2020-S2-QA-Q02	Unrated	From the interior point of an equilateral triangle, perpendiculars are drawn on ...	Open
3	24	CAT2020-S3-QA-Q24	Unrated	In a trepezium ABCD, AB is parallel to DC, BC is perpendicular ...	Open

CAT 2021 - 9 questions

1	6	CAT2021-S1-QA-Q06	Easy	If the area of a regular hexagon is equal to the area ...	Open
1	15	CAT2021-S1-QA-Q15	Difficult	Suppose the length of each side of a regular hexagon ABCDEF is ...	Open
2	9	CAT2021-S2-QA-Q09	Difficult	If a rhombus has area 12 sq cm and side length 5 ...	Open
2	11	CAT2021-S2-QA-Q11	Difficult	Let D and E be points on sides AB and AC, respectively, ...	Open
2	17	CAT2021-S2-QA-Q17	Difficult	The sides AB and CD of a trapezium ABCD are parallel, with ...	Open
3	5	CAT2021-S3-QA-Q05	Difficult	In a triangle ABC , □ BCA =50°. D and E are ...	Open
3	11	CAT2021-S3-QA-Q11	Difficult	Let ABCD be a parallelogram. The lengths of the side AD and ...	Open
3	12	CAT2021-S3-QA-Q12	Difficult	A park is shaped like a rhombus and has area 96 sq ...	Open
3	19	CAT2021-S3-QA-Q19	Difficult	The cost of fencing a rectangular plot is ₹ 200 per ft ...	Open

CAT 2022 - 5 questions

1	11	CAT2022-S1-QA-Q11	Unrated	A trapezium \(\text{ABCD}\) has side \(\text{AD}\) parallel to \(\text{BC}, \angle \text{BAD}=90^{\circ}, \text{BC}=3 ...	Open

2	1	CAT2022-S2-QA-Q01	Unrated	In triangle \(A B C\), altitudes \(A D\) and \(B E\) are ...	Open
2	21	CAT2022-S2-QA-Q21	Unrated	The length of each side of an equilateral triangle \(\mathrm{ABC}\) is \(3 ...	Open
3	7	CAT2022-S3-QA-Q07	Unrated	Suppose the medians BD and CE of a triangle ABC intersect at ...	Open
3	22	CAT2022-S3-QA-Q22	Unrated	The lengths of all four sides of a quadrilateral are integer valued. ...	Open

CAT 2023 - 6 questions

1	19	CAT2023-S1-QA-Q19	Unrated	In a right-angled triangle ABC, the altitude AB is 5 cm, and ...	Open
2	13	CAT2023-S2-QA-Q13	Unrated	The price of a precious stone is directly proportional to the square ...	Open
2	17	CAT2023-S2-QA-Q17	Unrated	In a rectangle ABCD, AB = 9 cm and BC = 6 ...	Open
2	19	CAT2023-S2-QA-Q19	Unrated	The area of the quadrilateral bounded by the \(Y\)-axis, the line \(x=5\), ...	Open
3	17	CAT2023-S3-QA-Q17	Unrated	Let \(\triangle A B C\) be an isosceles triangle such that \(A ...	Open
3	19	CAT2023-S3-QA-Q19	Unrated	In a regular polygon, any interior angle exceeds the exterior angle by ...	Open

CAT 2024 - 4 questions

1	2	CAT2024-S1-QA-Q02	Difficult	If \((a+b \sqrt{n})\) is the positive square root of \((29-12 \sqrt{5})\), where ...	Open
2	5	CAT2024-S2-QA-Q05	Difficult	\(A B C D\) is a trapezium in which \(A B\) is ...	Open
3	7	CAT2024-S3-QA-Q07	Difficult	The midpoints of sides AB, BC, and AC in □ ABC are ...	Open
3	21	CAT2024-S3-QA-Q21	Difficult	A regular octagon ABCDEFGH has sides of length 6 cm each. Then ...	Open

CAT 2025 - 6 questions

1	5	CAT2025-S1-QA-Q05	Difficult	In a 3-digit number N, the digits are non-zero and distinct such ...	Open
1	15	CAT2025-S1-QA-Q15	Difficult	If the length of a side of a rhombus is 36 cm ...	Open
2	21	CAT2025-S2-QA-Q21	Difficult	In a \(\triangle A B C \), points D and E ...	Open
3	1	CAT2025-S3-QA-Q01	Difficult	A triangle ABC is formed with AB = AC = 50 cm ...	Open
3	18	CAT2025-S3-QA-Q18	Difficult	In \(\triangle A B C , A B = A C ...	Open
3	21	CAT2025-S3-QA-Q21	Difficult	Let p, q and r be three natural numbers such that their ...	Open

ARITHMETIC

Averages

35 questions | 2017: 3 | 2018: 4 | 2019: 2 | 2020: 5 | 2021: 4 | 2022: 10 | 2023: 4 | 2024: 2 | 2025: 1

CAT 2017 - 3 questions

1	21	CAT2017-S1-QA-Q21	Unrated	Suppose, log 3 x = log 12 y = a, where x, ...	Open
2	6	CAT2017-S2-QA-Q06	Unrated	The average height of 22 toddlers increases by 2 inches when two ...	Open
2	26	CAT2017-S2-QA-Q26	Unrated	If log(2 a × 3 b × 5 c) is the ...	Open

CAT 2018 - 4 questions

1	5	CAT2018-S1-QA-Q05	Unrated	A CAT aspirant appears for a certain number of tests. His average ...	Open
1	34	CAT2018-S1-QA-Q34	Unrated	In an apartment complex, the number of people aged 51 years and ...	Open
2	2	CAT2018-S2-QA-Q02	Unrated	Let a 1 , a 2 , ... , a 52 be ...	Open
2	29	CAT2018-S2-QA-Q29	Unrated	The arithmetic mean of x, y and z is 80, and that ...	Open

CAT 2019 - 2 questions

1	34	CAT2019-S1-QA-Q34	Unrated	Ramesh and Gautam are among 22 students who write an examination. Ramesh ...	Open
2	2	CAT2019-S2-QA-Q02	Unrated	The average of 30 integers is 5. Among these 30 integers, there ...	Open

CAT 2020 - 5 questions

1	8	CAT2020-S1-QA-Q08	Unrated	Let A, B and C be three positive integers such that the ...	Open
1	10	CAT2020-S1-QA-Q10	Unrated	The mean of all 4 digit even natural numbers of the form ...	Open
2	3	CAT2020-S2-QA-Q03	Unrated	In a group of 10 students, the mean of the lowest 9 ...	Open
3	6	CAT2020-S3-QA-Q06	Unrated	Dick is thrice as old as Tom and Harry is twice as ...	Open
3	17	CAT2020-S3-QA-Q17	Unrated	A batsman played n + 2 innings and got out on all ...	Open

CAT 2021 - 4 questions

1	7	CAT2021-S1-QA-Q07	Difficult	Onion is sold for 5 consecutive months at the rate of Rs ...	Open
1	16	CAT2021-S1-QA-Q16	Difficult	Suppose hospital A admitted 21 less Covid infected patients than hospital B, ...	Open
2	14	CAT2021-S2-QA-Q14	Difficult	In a football tournament, a player has played a certain number of ...	Open
3	7	CAT2021-S3-QA-Q07	Difficult	The arithmetic mean of scores of 25 students in an examination is ...	Open

CAT 2022 - 10 questions

1	7	CAT2022-S1-QA-Q07	Unrated	The average of three integers is 13 . When a natural number ...	Open

1	20	CAT2022-S1-QA-Q20	Unrated	The average weight of students in a class increases by 600 gm ...	Open
2	4	CAT2022-S2-QA-Q04	Unrated	If \(a\) and \(b\) are non-negative real numbers such that \(a+2 b=6\), ...	Open
2	5	CAT2022-S2-QA-Q05	Unrated	Manu earns ₹4000 per month and wants to save an average of ...	Open
2	8	CAT2022-S2-QA-Q08	Unrated	The average of a non-decreasing sequence of \(N\) numbers \(a_1, a_2, \ldots, ...	Open
2	16	CAT2022-S2-QA-Q16	Unrated	Five students, including Amit, appear for an examination in which possible marks ...	Open
3	5	CAT2022-S3-QA-Q05	Unrated	In an examination, the average marks of students in sections A and ...	Open
3	8	CAT2022-S3-QA-Q08	Unrated	The arithmetic mean of all the distinct numbers that can be obtained ...	Open
3	13	CAT2022-S3-QA-Q13	Unrated	Consider six distinct natural numbers such that the average of the two ...	Open
3	17	CAT2022-S3-QA-Q17	Unrated	The average of all 3-digit terms in the arithmetic progression 38, 55, ...	Open

CAT 2023 - 4 questions

1	8	CAT2023-S1-QA-Q08	Unrated	Brishti went on an 8-hour trip in a car. Before the trip, ...	Open
1	10	CAT2023-S1-QA-Q10	Unrated	In an examination, the average marks of 4 girls and 6 boys ...	Open
1	20	CAT2023-S1-QA-Q20	Unrated	For some positive and distinct real numbers \(x, y\) and \(z\), if ...	Open
3	12	CAT2023-S3-QA-Q12	Unrated	There are three persons \(\mathrm{A}, \mathrm{B}\) and \(\mathrm{C}\) in a room. If ...	Open

CAT 2024 - 2 questions

1	11	CAT2024-S1-QA-Q11	Difficult	There are four numbers such that average of first two numbers is ...	Open
3	13	CAT2024-S3-QA-Q13	Difficult	The average of three distinct real numbers is 28. If the smallest ...	Open

CAT 2025 - 1 questions

2	3	CAT2025-S2-QA-Q03	Moderate	The average number of copies of a book sold per day by ...	Open

ARITHMETIC

Profit, Loss & Discount

33 questions | 2017: 6 | 2018: 3 | 2019: 3 | 2020: 3 | 2021: 6 | 2022: 2 | 2023: 4 | 2024: 3 | 2025: 3

CAT 2017 - 6 questions

1	6	CAT2017-S1-QA-Q06	Unrated	If a seller gives a discount of 15% on retail price, she ...	Open
1	8	CAT2017-S1-QA-Q08	Unrated	Suppose, C1, C2, C3, C4, and C5 are five companies. The profits ...	Open
1	11	CAT2017-S1-QA-Q11	Unrated	In a market, the price of medium quality mangoes is half that ...	Open
1	12	CAT2017-S1-QA-Q12	Unrated	If Fatima sells 60 identical toys at a 40% discount on the ...	Open
2	7	CAT2017-S2-QA-Q07	Unrated	The manufacturer of a table sells it to a wholesale dealer at ...	Open
2	9	CAT2017-S2-QA-Q09	Unrated	Mayank buys some candies for Rs 15 a dozen and an equal ...	Open

CAT 2018 - 3 questions

1	1	CAT2018-S1-QA-Q01	Unrated	A trader sells 10 litres of a mixture of paints A and ...	Open
1	6	CAT2018-S1-QA-Q06	Unrated	Two types of tea, A and B, are mixed and then sold ...	Open
1	7	CAT2018-S1-QA-Q07	Unrated	A wholesaler bought walnuts and peanuts, the price of walnut per kg ...	Open

CAT 2019 - 3 questions

1	13	CAT2019-S1-QA-Q13	Unrated	On selling a pen at 5% loss and a book at 15% ...	Open
2	27	CAT2019-S2-QA-Q27	Unrated	Mukesh purchased 10 bicycles in 2017, all at the same price. He ...	Open
2	33	CAT2019-S2-QA-Q33	Unrated	A shopkeeper sells two tables, each procured at cost price p, to ...	Open

CAT 2020 - 3 questions

1	12	CAT2020-S1-QA-Q12	Unrated	A person spent Rs 50000 to purchase a desktop computer and a ...	Open
2	11	CAT2020-S2-QA-Q11	Unrated	Anil buys 12 toys and labels each with the same selling price. ...	Open
3	14	CAT2020-S3-QA-Q14	Unrated	A man buys 35 kg of sugar and sets a marked price ...	Open

CAT 2021 - 6 questions

1	3	CAT2021-S1-QA-Q03	Difficult	Identical chocolate pieces are sold in boxes of two sizes, small and ...	Open
1	9	CAT2021-S1-QA-Q09	Difficult	Amal purchases some pens at ₹ 8 each. To sell these, he ...	Open
2	8	CAT2021-S2-QA-Q08	Difficult	Anil, Bobby and Chintu jointly invest in a business and agree to ...	Open
2	18	CAT2021-S2-QA-Q18	Difficult	Raj invested ₹ 10000 in a fund. At the end of first ...	Open
2	22	CAT2021-S2-QA-Q22	Moderate	A person buys tea of three different qualities at ₹ 800, ₹ ...	Open
3	8	CAT2021-S3-QA-Q08	Difficult	One part of a hostel's monthly expenses is fixed, and the other ...	Open

CAT 2022 - 2 questions

1	9	CAT2022-S1-QA-Q09	Unrated	Ankita buys 4 kg cashews, 14 kg peanuts and 6 kg almonds ...	Open
1	14	CAT2022-S1-QA-Q14	Unrated	Amal buys 110 kg of syrup and 120 kg of juice, syrup ...	Open

CAT 2023 - 4 questions

1	13	CAT2023-S1-QA-Q13	Unrated	Gita sells two objects A and B at the same price such ...	Open
2	9	CAT2023-S2-QA-Q09	Unrated	Minu purchases a pair of sunglasses at Rs.1000 and sells to Kanu ...	Open
2	15	CAT2023-S2-QA-Q15	Unrated	Jayant bought a certain number of white shirts at the rate of ...	Open
3	11	CAT2023-S3-QA-Q11	Unrated	A merchant purchases a cloth at a rate of Rs.100 per meter ...	Open

CAT 2024 - 3 questions

1	21	CAT2024-S1-QA-Q21	Moderate	The selling price of a product is fixed to ensure 40% profit. ...	Open
2	7	CAT2024-S2-QA-Q07	Moderate	Bina incurs 19% loss when she sells a product at Rs. 4860 ...	Open
3	12	CAT2024-S3-QA-Q12	Easy	Gopi marks a price on a product in order to make 20% ...	Open

CAT 2025 - 3 questions

1	16	CAT2025-S1-QA-Q16	Difficult	A shopkeeper offers a discount of 22% on the marked price of ...	Open
2	15	CAT2025-S2-QA-Q15	Difficult	An item with a cost price of Rs. 1650 is sold at ...	Open
3	5	CAT2025-S3-QA-Q05	Difficult	The monthly sales of a product from January to April were 120, ...	Open

ALGEBRA

Functions

28 questions | 2017: 3 | 2018: 2 | 2019: 2 | 2020: 3 | 2021: 3 | 2022: 3 | 2023: 2 | 2024: 5 | 2025: 5

CAT 2017 - 3 questions

1	33	CAT2017-S1-QA-Q33	Unrated	If f(x) = \\frac{5x + 2}{3x - 5}) and g(x) = x ...	Open
2	23	CAT2017-S2-QA-Q23	Unrated	Let f(x) = x 2 and g(x) = 2 x , for ...	Open
2	32	CAT2017-S2-QA-Q32	Unrated	Let f(x) = 2x – 5 and g(x) = 7 – 2x. ...	Open

CAT 2018 - 2 questions

1	33	CAT2018-S1-QA-Q33	Unrated	Let f(x)=min{2x 2 , 52 - 5x}, where x is any positive ...	Open
2	30	CAT2018-S2-QA-Q30	Unrated	Let f(x)=max{5x, 52 - 2x 2 }, where x is any positive ...	Open

CAT 2019 - 2 questions

1	2	CAT2019-S1-QA-Q02	Unrated	If (5.55) x = (0.555) y = 1000, then the value of ...	Open
2	29	CAT2019-S2-QA-Q29	Unrated	If (2n+1) + (2n+3) + (2n+5) + ... + (2n+47) = 5280 ...	Open

CAT 2020 - 3 questions

1	2	CAT2020-S1-QA-Q02	Unrated	If f(5 + x) = f(5 - x) for every real x ...	Open
2	14	CAT2020-S2-QA-Q14	Unrated	Let f(x) = x 2 + ax + b and g(x) = ...	Open
3	9	CAT2020-S3-QA-Q09	Unrated	If f(x+y) = f(x)f(y) and f(5) = 4, then f(10) - f(-10) ...	Open

CAT 2021 - 3 questions

1	1	CAT2021-S1-QA-Q01	Moderate	f(x) = \\frac{x^{2}+2 x-15}{x^{2}-7 x-18}) is negative if and only if	Open
2	13	CAT2021-S2-QA-Q13	Difficult	For all real values of x, the range of the function f(x) ...	Open
3	22	CAT2021-S3-QA-Q22	Moderate	If \(f(x)=x^{2}-7 x\) and \(g(x)=x+3\), then the minimum value of \(f(g(x))-3 x\) ...	Open

CAT 2022 - 3 questions

1	22	CAT2022-S1-QA-Q22	Unrated	Let \(0 \leq a \leq x \leq 100\) and \(f(x)=|x-a|+|x-100|+|x-a-50|\). Then the ...	Open
2	20	CAT2022-S2-QA-Q20	Unrated	Let \(f(x)\) be a quadratic polynomial in \(x\) such that \(f(x) \geq ...	Open
3	6	CAT2022-S3-QA-Q06	Unrated	Let \(r\) be a real number and \(f(x)=\left\{\begin{array}{cl}2 x-r & \text { ...	Open

CAT 2023 - 2 questions

3	5	CAT2023-S3-QA-Q05	Unrated	A quadratic equation \(x^2+b x+c=0\) has two real roots. If the difference ...	Open
3	22	CAT2023-S3-QA-Q22	Unrated	Suppose \(f(x, y)\) is a real-valued function such that \(f(3 x+2 y, ...	Open

CAT 2024 - 5 questions

1	7	CAT2024-S1-QA-Q07	Difficult	Consider two sets \(A=\{2,3,5,7,11,13\}\) and \(B=\{1,8,27\}\). Let \(f\) be a function from ...	Open
1	20	CAT2024-S1-QA-Q20	Difficult	If the equations \(x^2+m x+9=0, x^2+n x+17=0\) and \(x^2+(m+n) x+35=0\) have a ...	Open
2	13	CAT2024-S2-QA-Q13	Difficult	If \((x+6 \sqrt{2})^{\frac{1}{2}}-(x-6 \sqrt{2})^{\frac{1}{2}}=2 \sqrt{2}\), then \(x\) equals	Open
2	21	CAT2024-S2-QA-Q21	Difficult	If \(x\) and \(y\) are real numbers such that \(4 x^2+4 y^2-4 ...	Open
3	20	CAT2024-S3-QA-Q20	Difficult	For any non-zero real number \(x\), let \(f(x)+2 f\left(\frac{1}{x}\right)=3 x\). Then, the ...	Open

CAT 2025 - 5 questions

1	8	CAT2025-S1-QA-Q08	Difficult	A value of \(c\) for which the minimum value of \(f(x)=x^2-4 c ...	Open
2	2	CAT2025-S2-QA-Q02	Difficult	If \(\log _ { 64 } x ^ { 2 } ...	Open
2	4	CAT2025-S2-QA-Q04	Difficult	Let \(f (x) = \frac { x } { ...	Open
3	11	CAT2025-S3-QA-Q11	Difficult	If \(f (x) = \left(x ^ { 2 ...	Open
3	15	CAT2025-S3-QA-Q15	Difficult	For real values of \(x \), the range of the function ...	Open

ALGEBRA

Sequences & Series

22 questions | 2017: 4 | 2018: 1 | 2019: 3 | 2020: 1 | 2021: 2 | 2022: 3 | 2023: 1 | 2024: 4 | 2025: 3

CAT 2017 - 4 questions

1	3	CAT2017-S1-QA-Q03	Unrated	An elevator has a weight limit of 630 kg. It is carrying ...	Open
2	27	CAT2017-S2-QA-Q27	Unrated	Let a 1 , a 2 , a 3 , a 4 ...	Open
2	33	CAT2017-S2-QA-Q33	Unrated	An infinite geometric progression a 1 , a 2 , a 3 ...	Open
2	34	CAT2017-S2-QA-Q34	Unrated	If a 1 = \\frac{1}{2 × 5}) , a 2 = \\frac{1}{5 ...	Open

CAT 2018 - 1 questions

2	16	CAT2018-S2-QA-Q16	Unrated	The value of the sum 7 x 11 + 11 x 15 ...	Open

CAT 2019 - 3 questions

1	16	CAT2019-S1-QA-Q16	Unrated	If a 1 + a 2 + a 3 + ... + ...	Open
1	20	CAT2019-S1-QA-Q20	Unrated	If a 1 , a 2 , ... are in A.P , ...	Open
2	28	CAT2019-S2-QA-Q28	Unrated	The number of common terms in the two sequences: 15, 19, 23, ...	Open

CAT 2020 - 1 questions

1	25	CAT2020-S1-QA-Q25	Unrated	A gentleman decided to treat a few children in the following manner. ...	Open

CAT 2021 - 2 questions

1	22	CAT2021-S1-QA-Q22	Difficult	If x 0 = 1, x 1 = 2, and x n ...	Open
3	16	CAT2021-S3-QA-Q16	Difficult	Consider a sequence of real numbers \(x_{1}, x_{2}, x_{3}, \ldots\) such that ...	Open

CAT 2022 - 3 questions

1	10	CAT2022-S1-QA-Q10	Unrated	The largest real value of \(a\) for which the equation \(|x+a|+|x-1|=2\) has ...	Open
2	15	CAT2022-S2-QA-Q15	Unrated	In an examination, there were 75 questions. 3 marks were awarded for ...	Open
2	19	CAT2022-S2-QA-Q19	Unrated	Consider the arithmetic progression \(3,7,11, \ldots\) and let \(A_n\) denote the sum ...	Open

CAT 2023 - 1 questions

3	1	CAT2023-S3-QA-Q01	Unrated	For a real number \(x\), if \(\frac{1}{2}, \frac{\log _3\left(2^x-9\right)}{\log _3 4}\), and ...	Open

CAT 2024 - 4 questions

1	13	CAT2024-S1-QA-Q13	Moderate	A shop wants to sell a certain quantity (in kg) of grains. ...	Open
1	14	CAT2024-S1-QA-Q14	Moderate	Suppose \(x_1, x_2, x_3, \ldots, x_{100}\) are in arithmetic progression such that ...	Open
2	8	CAT2024-S2-QA-Q08	Difficult	The sum of the infinite series \(\frac{1}{5}\left(\frac{1}{5}-\frac{1}{7}\right)+\left(\frac{1}{5}\right)^2\left(\left(\frac{1}{5}\right)^2-\left(\frac{1}{7}\right)^2\right)+\left(\frac{1}{5}\right)^3\left(\left(\frac{1}{5}\right)^3-\left(\frac{1}{7}\right)^3\right)+\cdots\) is equal to	Open
3	17	CAT2024-S3-QA-Q17	Difficult	Consider the sequence \(t_1=1, t_2=-1\) and \(t_n=\left(\frac{n-3}{n-1}\right) t_{n-2}\) for \(n \geq 3\). ...	Open

CAT 2025 - 3 questions

1	2	CAT2025-S1-QA-Q02	Difficult	A cafeteria offers 5 types of sandwiches. Moreover, for each type of ...	Open
2	8	CAT2025-S2-QA-Q08	Difficult	Let \(a _ { n } \) be the \(n ...	Open
3	2	CAT2025-S3-QA-Q02	Moderate	In an arithmetic progression, if the sum of fourth, seventh and tenth ...	Open

GEOMETRY

Circles

20 questions | 2017: 2 | 2018: 4 | 2019: 3 | 2020: 2 | 2021: 1 | 2022: 2 | 2023: 2 | 2024: 2 | 2025: 2

CAT 2017 - 2 questions

1	17	CAT2017-S1-QA-Q17	Unrated	Let ABC be a right-angled isosceles triangle with hypotenuse BC. Let BQC ...	Open
2	18	CAT2017-S2-QA-Q18	Unrated	ABCD is a quadrilateral inscribed in a circle with centre O. If ...	Open

CAT 2018 - 4 questions

1	2	CAT2018-S1-QA-Q02	Unrated	In a circle with centre O and radius 1 cm, an arc ...	Open
1	18	CAT2018-S1-QA-Q18	Unrated	Let ABCD be a rectangle inscribed in a circle of radius 13 ...	Open
2	4	CAT2018-S2-QA-Q04	Unrated	On a triangle ABC, a circle with diameter BC is drawn, intersecting ...	Open
2	8	CAT2018-S2-QA-Q08	Unrated	A chord of length 5 cm subtends an angle of 60° at ...	Open

CAT 2019 - 3 questions

1	5	CAT2019-S1-QA-Q05	Unrated	In a circle of radius 11 cm, CD is a diameter and ...	Open
1	21	CAT2019-S1-QA-Q21	Unrated	AB is a diameter of a circle of radius 5 cm. Let ...	Open
2	7	CAT2019-S2-QA-Q07	Unrated	Two circles, each of radius 4 cm, touch externally. Each of these ...	Open

CAT 2020 - 2 questions

2	9	CAT2020-S2-QA-Q09	Unrated	Let C be a circle of radius 5 meters having center at ...	Open
2	21	CAT2020-S2-QA-Q21	Unrated	Let C1 and C2 be concentric circles such that the diameter of ...	Open

CAT 2021 - 1 questions

1	14	CAT2021-S1-QA-Q14	Difficult	A circle of diameter 8 inches is inscribed in a triangle ABC ...	Open

CAT 2022 - 2 questions

1	19	CAT2022-S1-QA-Q19	Unrated	All the vertices of a rectangle lie on a circle of radius ...	Open
3	18	CAT2022-S3-QA-Q18	Unrated	In a triangle $\mathrm{ABC}, \mathrm{AB}=\mathrm{AC}=8 \mathrm{cm}$. A circle drawn with BC as ...	Open

CAT 2023 - 2 questions

1	17	CAT2023-S1-QA-Q17	Unrated	Let C be the circle $x^2+y^2+4 x-6 y-3=0$ and L be the ...	Open
1	18	CAT2023-S1-QA-Q18	Unrated	A quadrilateral ABCD is inscribed in a circle such that AB : ...	Open

CAT 2024 - 2 questions

1	19	CAT2024-S1-QA-Q19	Difficult	ABCD is a rectangle with sides AB = 56 cm and BC ...	Open
3	4	CAT2024-S3-QA-Q04	Difficult	A circular plot of land is divided into two regions by a ...	Open

CAT 2025 - 2 questions

2	10	CAT2025-S2-QA-Q10	Difficult	Two tangents drawn from a point \(P \) touch a circle ...	Open
3	3	CAT2025-S3-QA-Q03	Difficult	ABCD is a trapezium in which AB is parallel to \(\mathrm ...	Open

ARITHMETIC

Simple & Compound Interest

19 questions | 2018: 2 | 2019: 2 | 2020: 3 | 2021: 2 | 2022: 3 | 2023: 2 | 2024: 3 | 2025: 2

CAT 2018 - 2 questions

1	15	CAT2018-S1-QA-Q15	Unrated	John borrowed Rs. 2,10,000 from a bank at an interest rate of ...	Open
2	28	CAT2018-S2-QA-Q28	Unrated	Gopal borrows Rs. X from Ankit at 8% annual interest. He then ...	Open

CAT 2019 - 2 questions

1	11	CAT2019-S1-QA-Q11	Unrated	A person invested a total amount of Rs 15 lakh. A part ...	Open
2	32	CAT2019-S2-QA-Q32	Unrated	Amal invests Rs 12000 at 8% interest, compounded annually, and Rs 10000 ...	Open

CAT 2020 - 3 questions

1	3	CAT2020-S1-QA-Q03	Unrated	Veeru invested Rs 10000 at 5% simple annual interest, and exactly after ...	Open
2	8	CAT2020-S2-QA-Q08	Unrated	For the same principal amount, the compound interest for two years at ...	Open
3	20	CAT2020-S3-QA-Q20	Unrated	A person invested a certain amount of money at 10% annual interest, ...	Open

CAT 2021 - 2 questions

1	10	CAT2021-S1-QA-Q10	Moderate	Anil invests some money at a fixed rate of interest, compounded annually. ...	Open
3	21	CAT2021-S3-QA-Q21	Difficult	Bank A offers 6% interest rate per annum compounded half yearly. Bank ...	Open

CAT 2022 - 3 questions

1	16	CAT2022-S1-QA-Q16	Unrated	Alex invested his savings in two parts. The simple interest earned on ...	Open
2	18	CAT2022-S2-QA-Q18	Unrated	Mr. Pinto invests one-fifth of his capital at 6%, one-third at 10% ...	Open
3	9	CAT2022-S3-QA-Q09	Unrated	Nitu has an initial capital of \(₹ 20,000\). Out of this, she ...	Open

CAT 2023 - 2 questions

1	16	CAT2023-S1-QA-Q16	Unrated	Anil invests Rs. 22000 for 6 years in a certain scheme with ...	Open
2	12	CAT2023-S2-QA-Q12	Unrated	Anil borrows Rs 2 lakhs at an interest rate of 8% per ...	Open

CAT 2024 - 3 questions

1	12	CAT2024-S1-QA-Q12	Easy	An amount of Rs 10000 is deposited in bank A for a ...	Open
2	15	CAT2024-S2-QA-Q15	Moderate	Anil invests Rs 22000 for 6 years in a scheme with 4% ...	Open
3	11	CAT2024-S3-QA-Q11	Difficult	Aman invests Rs 4000 in a bank at a certain rate of ...	Open

CAT 2025 - 2 questions

1	6	CAT2025-S1-QA-Q06	Difficult	At a certain simple rate of interest, a given sum amounts to ...	Open
2	18	CAT2025-S2-QA-Q18	Moderate	A loan of Rs 1000 is fully repaid by two installments of ...	Open

ARITHMETIC

Time & Work

14 questions | 2018: 1 | 2019: 2 | 2020: 2 | 2021: 4 | 2022: 1 | 2023: 2 | 2025: 2

CAT 2018 - 1 questions

1	28	CAT2018-S1-QA-Q28	Unrated	Humans and robots can both perform a job but at different efficiencies. ...	Open

CAT 2019 - 2 questions

1	15	CAT2019-S1-QA-Q15	Unrated	At their usual efficiency levels, A and B together finish a task ...	Open
1	30	CAT2019-S1-QA-Q30	Unrated	Three men and eight machines can finish a job in half the ...	Open

CAT 2020 - 2 questions

2	22	CAT2020-S2-QA-Q22	Unrated	John takes twice as much time as Jack to finish a job. ...	Open
3	4	CAT2020-S3-QA-Q04	Unrated	A contractor agreed to construct a 6 km road in 200 days. ...	Open

CAT 2021 - 4 questions

1	12	CAT2021-S1-QA-Q12	Moderate	Amar, Akbar and Anthony are working on a project. Working together Amar ...	Open
1	19	CAT2021-S1-QA-Q19	Easy	Anu, Vinu and Manu can complete a work alone in 15 days, ...	Open
2	6	CAT2021-S2-QA-Q06	Moderate	Anil can paint a house in 60 days while Bimal can paint ...	Open
3	9	CAT2021-S3-QA-Q09	Difficult	One day, Rahul started a work at 9 AM and Gautam joined ...	Open

CAT 2022 - 1 questions

3	3	CAT2022-S3-QA-Q03	Unrated	Bob can finish a job in 40 days, if he works alone. ...	Open

CAT 2023 - 2 questions

1	15	CAT2023-S1-QA-Q15	Unrated	The amount of job that Amal, Sunil and Kamal can individually do ...	Open
3	8	CAT2023-S3-QA-Q08	Unrated	Rahul, Rakshita and Gurmeet, working together, would have taken more than 7 ...	Open

CAT 2025 - 2 questions

1	18	CAT2025-S1-QA-Q18	Difficult	Arun, Varun and Tarun, if working alone, can complete a task in ...	Open
2	12	CAT2025-S2-QA-Q12	Moderate	Ankita is twice as efficient as Bipin, while Bipin is twice as ...	Open

MODERN MATH

Permutations & Combinations

14 questions | 2017: 5 | 2018: 1 | 2020: 1 | 2021: 2 | 2022: 1 | 2023: 1 | 2024: 3

CAT 2017 - 5 questions

1	25	CAT2017-S1-QA-Q25	Unrated	The number of solutions (x, y, z) to the equation x – ...	Open
1	29	CAT2017-S1-QA-Q29	Unrated	Let AB, CD, EF, GH, and JK be five diameters of a ...	Open
1	32	CAT2017-S1-QA-Q32	Unrated	In how many ways can 7 identical erasers be distributed among 4 ...	Open
2	29	CAT2017-S2-QA-Q29	Unrated	In how many ways can 8 identical pens be distributed among Amal, ...	Open
2	30	CAT2017-S2-QA-Q30	Unrated	How many four digit numbers, which are divisible by 6, can be ...	Open

CAT 2018 - 1 questions

2	10	CAT2018-S2-QA-Q10	Unrated	In a tournament, there are 43 junior level and 51 senior level ...	Open

CAT 2020 - 1 questions

3	16	CAT2020-S3-QA-Q16	Unrated	How many integers in the set {100, 101, 102, ..., 999} have ...	Open

CAT 2021 - 2 questions

1	13	CAT2021-S1-QA-Q13	Difficult	The number of groups of three or more distinct numbers that can ...	Open
2	1	CAT2021-S2-QA-Q01	Difficult	The number of ways of distributing 15 identical balloons, 6 identical pencils ...	Open

CAT 2022 - 1 questions

1	2	CAT2022-S1-QA-Q02	Unrated	The number of ways of distributing 20 identical balloons among 4 children ...	Open

CAT 2023 - 1 questions

1	21	CAT2023-S1-QA-Q21	Unrated	The number of all natural numbers up to 1000 with non-repeating digits ...	Open

CAT 2024 - 3 questions

1	9	CAT2024-S1-QA-Q09	Difficult	The sum of all four-digit numbers that can be formed with the ...	Open
2	11	CAT2024-S2-QA-Q11	Difficult	P, Q, R and S are four towns. One can travel between ...	Open
3	14	CAT2024-S3-QA-Q14	Difficult	The number of all positive integers up to 500 with non-repeating digits ...	Open

ALGEBRA

Linear & Quadratic Equations

12 questions | 2017: 1 | 2018: 1 | 2019: 2 | 2020: 2 | 2021: 3 | 2022: 1 | 2023: 1 | 2024: 1

CAT 2017 - 1 questions

1	22	CAT2017-S1-QA-Q22	Unrated	If x + 1 = x 2 and x > 0, then ...	Open

CAT 2018 - 1 questions

1	16	CAT2018-S1-QA-Q16	Unrated	If u 2 + (u−2v−1) 2 = −4v(u + v), then what ...	Open

CAT 2019 - 2 questions

1	29	CAT2019-S1-QA-Q29	Unrated	The number of solutions of the equation |x|(6x 2 + 1) = ...	Open
1	31	CAT2019-S1-QA-Q31	Unrated	The product of the distinct roots of |x 2 - x - ...	Open

CAT 2020 - 2 questions

1	11	CAT2020-S1-QA-Q11	Unrated	The number of distinct real roots of the equation (x + \\frac{1}{x})) ...	Open
3	15	CAT2020-S3-QA-Q15	Unrated	Let k be a constant. The equations kx + y = 3 ...	Open

CAT 2021 - 3 questions

1	8	CAT2021-S1-QA-Q08	Difficult	If r is a constant such that |x 2 - 4 x ...	Open
2	3	CAT2021-S2-QA-Q03	Difficult	Consider the pair of equations: x 2 - xy - x = ...	Open
2	16	CAT2021-S2-QA-Q16	Difficult	For all real numbers x the condition |3x - 20| + |3x ...	Open

CAT 2022 - 1 questions

3	15	CAT2022-S3-QA-Q15	Unrated	If \(c=\frac{16 x}{y}+\frac{49 y}{x}\) for some non-zero real numbers \(x\) and \(y\), ...	Open

CAT 2023 - 1 questions

2	4	CAT2023-S2-QA-Q04	Unrated	Any non-zero real numbers \(x, y\) such that \(y \neq 3\) and ...	Open

CAT 2024 - 1 questions

3	2	CAT2024-S3-QA-Q02	Difficult	For some constant real numbers \(p, k\) and \(a\), consider the following ...	Open

ARITHMETIC

Mixtures & Alligation

12 questions | 2017: 3 | 2018: 1 | 2020: 1 | 2021: 1 | 2022: 3 | 2023: 2 | 2024: 1

CAT 2017 - 3 questions

1	14	CAT2017-S1-QA-Q14	Unrated	A class consists of 20 boys and 30 girls. In the mid-semester ...	Open
2	3	CAT2017-S2-QA-Q03	Unrated	Bottle 1 contains a mixture of milk and water in 7 : ...	Open
2	14	CAT2017-S2-QA-Q14	Unrated	Consider three mixtures - the first having water and liquid A in ...	Open

CAT 2018 - 1 questions

2	3	CAT2018-S2-QA-Q03	Unrated	There are two drums, each containing a mixture of paints A and ...	Open

CAT 2020 - 1 questions

1	26	CAT2020-S1-QA-Q26	Unrated	A solution, of volume 40 litres, has dye and water in the ...	Open

CAT 2021 - 1 questions

2	4	CAT2021-S2-QA-Q04	Difficult	From a container filled with milk, 9 litres of milk are drawn ...	Open

CAT 2022 - 3 questions

1	3	CAT2022-S1-QA-Q03	Unrated	A mixture contains lemon juice and sugar syrup in equal proportion. If ...	Open
2	6	CAT2022-S2-QA-Q06	Unrated	There are two containers of the same volume, first container half-filled with ...	Open
3	4	CAT2022-S3-QA-Q04	Unrated	A glass contains 500 cc of milk and a cup contains 500 ...	Open

CAT 2023 - 2 questions

1	11	CAT2023-S1-QA-Q11	Unrated	A mixture P is formed by removing a certain amount of coffee ...	Open
2	14	CAT2023-S2-QA-Q14	Unrated	A container has 40 liters of milk. Then, 4 liters are removed ...	Open

CAT 2024 - 1 questions

1	17	CAT2024-S1-QA-Q17	Difficult	A glass is filled with milk. Two-thirds of its content is poured ...	Open

GEOMETRY

Coordinate Geometry

12 questions | 2017: 2 | 2019: 3 | 2020: 4 | 2022: 1 | 2024: 1 | 2025: 1

CAT 2017 - 2 questions

1	15	CAT2017-S1-QA-Q15	Unrated	The area of the closed region bounded by the equation | x ...	Open
2	17	CAT2017-S2-QA-Q17	Unrated	The points (2, 5) and (6, 3) are two end points of ...	Open

CAT 2019 - 3 questions

1	3	CAT2019-S1-QA-Q03	Unrated	With rectangular axes of coordinates, the number of paths from (1,1) to ...	Open
1	8	CAT2019-S1-QA-Q08	Unrated	Let T be the triangle formed by the straight line 3x + ...	Open
1	28	CAT2019-S1-QA-Q28	Unrated	Let S be the set of all points (x,y) in the x-y ...	Open

CAT 2020 - 4 questions

1	16	CAT2020-S1-QA-Q16	Unrated	The area of the region satisfying the inequalities |x| - y ≤ ...	Open
3	5	CAT2020-S3-QA-Q05	Unrated	The area, in sq. units, enclosed by the lines x = 2, ...	Open
3	19	CAT2020-S3-QA-Q19	Unrated	The vertices of a triangle are (0,0), (4,0) and (3,9). The area ...	Open
3	25	CAT2020-S3-QA-Q25	Unrated	The points (2 , 1) and (-3 , -4) are opposite vertices ...	Open

CAT 2022 - 1 questions

1	1	CAT2022-S1-QA-Q01	Unrated	Let ABCD be a parallelogram such that the coordinates of its three ...	Open

CAT 2024 - 1 questions

2	14	CAT2024-S2-QA-Q14	Moderate	The coordinates of the three vertices of a triangle are: (1, 2), ...	Open

CAT 2025 - 1 questions

1	13	CAT2025-S1-QA-Q13	Difficult	The (x, y) coordinates of vertices P, Q and R of a ...	Open

OTHER

Mixed / Needs Review

12 questions | 2019: 1 | 2020: 1 | 2021: 3 | 2022: 1 | 2023: 2 | 2025: 4

CAT 2019 - 1 questions

1	10	CAT2019-S1-QA-Q10	Unrated	If the population of a town is p in the beginning of ...	Open

CAT 2020 - 1 questions

1	13	CAT2020-S1-QA-Q13	Unrated	Among 100 students, x 1 have birthdays in January, x 2 have ...	Open

CAT 2021 - 3 questions

1	5	CAT2021-S1-QA-Q05	Difficult	A basket of 2 apples, 4 oranges and 6 mangoes costs the ...	Open
3	13	CAT2021-S3-QA-Q13	Difficult	If $(3 x+2\|y\|+y=7)$ and $(x+\|x\|+3 y=1)$, then $(x+2 y)$ is	Open
3	15	CAT2021-S3-QA-Q15	Difficult	A shop owner bought a total of 64 shirts from a wholesale ...	Open

CAT 2022 - 1 questions

2	7	CAT2022-S2-QA-Q07	Unrated	On day one, there are 100 particles in a laboratory experiment. On ...	Open

CAT 2023 - 2 questions

1	3	CAT2023-S1-QA-Q03	Unrated	If $(\sqrt{5 x+9}+\sqrt{5 x-9}=3(2+\sqrt{2}))$, then $(\sqrt{10 x+9})$ is equal to	Open
3	20	CAT2023-S3-QA-Q20	Unrated	The value of $(1+\left(1+\frac{1}{3}\right) \frac{1}{4}+\left(1+\frac{1}{3}+\frac{1}{9}\right) \frac{1}{16}+\left(1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}\right) \frac{1}{64}+\cdots)$, is	Open

CAT 2025 - 4 questions

1	1	CAT2025-S1-QA-Q01	Difficult	If $(a-6 b+6 c=4)$ and $(6 a+3 b-3 c=50)$, where (a, b) ...	Open
1	4	CAT2025-S1-QA-Q04	Difficult	In the set of consecutive odd numbers $(\{1,3,5, \ldots, 57\})$, there is ...	Open
1	20	CAT2025-S1-QA-Q20	Easy	Stocks A, B and C are priced at rupees 120, 90 and ...	Open
3	10	CAT2025-S3-QA-Q10	Difficult	Teams A, B, and C consist of five, eight, and ten members, ...	Open

ALGEBRA

Inequalities & Optimization

11 questions | 2017: 1 | 2020: 2 | 2022: 2 | 2023: 1 | 2024: 3 | 2025: 2

CAT 2017 - 1 questions

2	24	CAT2017-S2-QA-Q24	Unrated	The minimum possible value of the sum of the squares of the ...	Open

CAT 2020 - 2 questions

2	7	CAT2020-S2-QA-Q07	Unrated	If x and y are positive real numbers satisfying x + y ...	Open
2	24	CAT2020-S2-QA-Q24	Unrated	Aron bought some pencils and sharpeners. Spending the same amount of money ...	Open

CAT 2022 - 2 questions

1	17	CAT2022-S1-QA-Q17	Unrated	In a class of 100 students, 73 like coffee, 80 like tea ...	Open
3	20	CAT2022-S3-QA-Q20	Unrated	A donation box can receive only cheques of ₹100, ₹250, and ₹500. ...	Open

CAT 2023 - 1 questions

2	16	CAT2023-S2-QA-Q16	Unrated	If a certain amount of money is divided equally among \(n\) persons, ...	Open

CAT 2024 - 3 questions

2	2	CAT2024-S2-QA-Q02	Difficult	If \(x\) and \(y\) satisfy the equations \(|x|+x+y=15\) and \(x+|y|-y=20\), then \((x-y)\) ...	Open
2	20	CAT2024-S2-QA-Q20	Moderate	All the values of \(x\) satisfying the inequality \(\frac{1}{x+5} \leq \frac{1}{2 x-3}\) ...	Open
3	18	CAT2024-S3-QA-Q18	Difficult	The number of distinct real values of \(x\), satisfying the equation \(\max ...	Open

CAT 2025 - 2 questions

3	13	CAT2025-S3-QA-Q13	Difficult	In a school with 1500 students, each student chooses any one of ...	Open
3	14	CAT2025-S3-QA-Q14	Difficult	In a class of 150 students, 75 students chose physics, 111 students ...	Open

NUMBER SYSTEM

Digits & Number Bases

11 questions | 2018: 2 | 2019: 1 | 2020: 2 | 2021: 3 | 2022: 1 | 2023: 1 | 2025: 1

CAT 2018 - 2 questions

1	14	CAT2018-S1-QA-Q14	Unrated	How many numbers with two or more digits can be formed with ...	Open
2	26	CAT2018-S2-QA-Q26	Unrated	How many two-digit numbers, with a non-zero digit in the units place, ...	Open

CAT 2019 - 1 questions

2	26	CAT2019-S2-QA-Q26	Unrated	In a six-digit number, the sixth, that is, the rightmost, digit is ...	Open

CAT 2020 - 2 questions

1	1	CAT2020-S1-QA-Q01	Unrated	How many 3-digit numbers are there, for which the product of their ...	Open
2	26	CAT2020-S2-QA-Q26	Unrated	How many 4-digit numbers, each greater than 1000 and each having all ...	Open

CAT 2021 - 3 questions

1	20	CAT2021-S1-QA-Q20	Difficult	How many three-digit numbers are greater than 100 and increase by 198 ...	Open
2	12	CAT2021-S2-QA-Q12	Difficult	For a 4-digit number, the sum of its digits in the thousands, ...	Open
3	4	CAT2021-S3-QA-Q04	Difficult	A four-digit number is formed by using only the digits 1, 2 ...	Open

CAT 2022 - 1 questions

2	12	CAT2022-S2-QA-Q12	Unrated	The number of integers greater than 2000 that can be formed with ...	Open

CAT 2023 - 1 questions

2	22	CAT2023-S2-QA-Q22	Unrated	Let \(a_n\) and \(b_n\) be two sequences such that \(a_n=13+6(n-1)\) and \(b_n=15+7(n-1)\) ...	Open

CAT 2025 - 1 questions

3	20	CAT2025-S3-QA-Q20	Difficult	For a 4-digit number (greater than 1000), sum of the digits in ...	Open

ARITHMETIC

Pipes & Cisterns

10 questions | 2017: 2 | 2018: 3 | 2021: 1 | 2023: 1 | 2024: 3

CAT 2017 - 2 questions

1	2	CAT2017-S1-QA-Q02	Unrated	A person can complete a job in 120 days. He works alone ...	Open
2	8	CAT2017-S2-QA-Q08	Unrated	A tank has an inlet pipe and an outlet pipe. If the ...	Open

CAT 2018 - 3 questions

1	31	CAT2018-S1-QA-Q31	Unrated	A tank is fitted with pipes, some filling it and the rest ...	Open
2	21	CAT2018-S2-QA-Q21	Unrated	A tank is emptied everyday at a fixed time point. Immediately thereafter, ...	Open
2	27	CAT2018-S2-QA-Q27	Unrated	A water tank has inlets of two types A and B. All ...	Open

CAT 2021 - 1 questions

2	21	CAT2021-S2-QA-Q21	Moderate	Two pipes A and B are attached to an empty water tank. ...	Open

CAT 2023 - 1 questions

2	10	CAT2023-S2-QA-Q10	Unrated	Pipes A and C are fill pipes while Pipe B is a ...	Open

CAT 2024 - 3 questions

1	5	CAT2024-S1-QA-Q05	Difficult	Renu would take 15 days working 4 hours per day to complete ...	Open
2	6	CAT2024-S2-QA-Q06	Difficult	Amal and Vimal together can complete a task in 150 days, while ...	Open
3	9	CAT2024-S3-QA-Q09	Easy	Sam can complete a job in 20 days when working alone. Mohit ...	Open

GEOMETRY

Mensuration

9 questions | 2018: 4 | 2019: 2 | 2020: 2 | 2024: 1

CAT 2018 - 4 questions

1	13	CAT2018-S1-QA-Q13	Unrated	A right circular cone, of height 12 ft, stands on its base ...	Open
1	29	CAT2018-S1-QA-Q29	Unrated	In a parallelogram ABCD of area 72 sq cm, the sides CD ...	Open
2	6	CAT2018-S2-QA-Q06	Unrated	From a rectangle ABCD of area 768 sq cm, a semicircular part ...	Open
2	15	CAT2018-S2-QA-Q15	Unrated	A parallelogram ABCD has area 48 sqcm. If the length of CD ...	Open

CAT 2019 - 2 questions

2	18	CAT2019-S2-QA-Q18	Unrated	The base of a regular pyramid is a square and each of ...	Open
2	24	CAT2019-S2-QA-Q24	Unrated	A man makes complete use of 405 cc of iron, 783 cc ...	Open

CAT 2020 - 2 questions

1	17	CAT2020-S1-QA-Q17	Unrated	A solid right circular cone of height 27 cm is cut into ...	Open
1	22	CAT2020-S1-QA-Q22	Unrated	On a rectangular metal sheet of area 135 sq in, a circle ...	Open

CAT 2024 - 1 questions

1	16	CAT2024-S1-QA-Q16	Difficult	The surface area of a closed rectangular box, which is inscribed in ...	Open

MODERN MATH

Set Theory

6 questions | 2018: 4 | 2019: 1 | 2020: 1

CAT 2018 - 4 questions

1	22	CAT2018-S1-QA-Q22	Unrated	Each of 74 students in a class studies at least one of ...	Open
1	32	CAT2018-S1-QA-Q32	Unrated	If among 200 students, 105 like pizza and 134 like burger, then ...	Open
2	31	CAT2018-S2-QA-Q31	Unrated	For two sets A and B, let AΔB denote the set of ...	Open
2	32	CAT2018-S2-QA-Q32	Unrated	If A = {6 2n - 35n - 1: n = 1,2,3,...} ...	Open

CAT 2019 - 1 questions

1	4	CAT2019-S1-QA-Q04	Unrated	A club has 256 members of whom 144 can play football, 123 ...	Open

CAT 2020 - 1 questions

2	13	CAT2020-S2-QA-Q13	Unrated	Students in a college have to choose at least two subjects from ...	Open

NUMBER SYSTEM

Remainders

3 questions | 2024: 3

CAT 2024 - 3 questions

1	10	CAT2024-S1-QA-Q10	Moderate	When \(10^{100}\) is divided by 7 , the remainder is	Open
2	1	CAT2024-S2-QA-Q01	Moderate	When \(3^{333}\) is divided by 11 , the remainder is	Open
3	6	CAT2024-S3-QA-Q06	Moderate	If \(10^{68}\) is divided by 13 , the remainder is	Open

GEOMETRY

Trigonometry

1 questions | 2019: 1

CAT 2019 - 1 questions

1	17	CAT2019-S1-QA-Q17	Unrated	The number of the real roots of the equation 2cos(x(x + 1)) ...	Open

NUMBER SYSTEM

Factorials

1 questions | 2022: 1

CAT 2022 - 1 questions

2	**13**	CAT2022-S2-QA-Q13	Unrated	For some natural number \(n\), assume that \((15,000)\) ! is divisible by ...	Open

NUMBER SYSTEM

Fractions & Decimals

1 questions | 2021: 1

CAT 2021 - 1 questions

2	**19**	CAT2021-S2-QA-Q19	Difficult	Suppose one of the roots of the equation a x 2 - ...	Open